THE PICARD GROUP OF THE GENERIC FIBRE VANISHES FOR A FLAT ESSENTIALLY FINITE TYPE LOCAL ALGEBRA OVER A VALUATION RING WITH REGULAR SPECIAL FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span contains the author's three enumerated hypotheses, delivered verbatim. The author uses italic markup for the word 'free' inside the counted proof span; that is the author's text and is delivered as it stands. The seven references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Torsion free modules over a valuation ring are flat, and flat finite type modules are finitely presented. invoked at proof lines 36864-36868. Gives flatness and finite presentation of the descended module M.
- Relative perfectness and perfection over a regular local ring. invoked at proof lines 36869-36874. Upgrades M to a perfect module with a finite free resolution.
- The Euler characteristic on K_0 and the determinant map. invoked at proof lines 36877-36881. Turn the finite free resolution into the class statement and then into triviality of L.

Source reference. Original lemma 05N5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7294–7294. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 36464: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36464–36464. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 5967: sources/stacks-more-algebra/pinned-original-source.tex, source lines 5967–5967. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 20229: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20229–20229. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 36491: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36491–36491. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 21980: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21980–21980. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 4925: sources/stacks-more-algebra/pinned-original-source.tex, source lines 4925–4925. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a valuation ring with fraction field K and residue field κ. Let $R \to A$ be a homomorphism of rings such that*

(1) *A is local and $R \to A$ is local,*
(2) *A is flat and essentially of finite type over R,*
(3) *$A \otimes_R \kappa$ regular.*

Then $\mathrm{Pic}(A \otimes_R K) = 0$.

Proof. Let L be an invertible $A \otimes_R K$-module. In particular L is a finite module. There exists a finite A-module M such that $M \otimes_R K \cong L$, see Algebra, Lemma 05N5. We may assume M is torsion free as an R-module. Thus M is flat as an R-module (Lemma 4925). From Lemma 5967 we deduce that M is of finite presentation as an A-module and A is essentially of finite presentation as an R-algebra. By Lemma 21980 we see that M is perfect relative to R, in particular M is pseudo-coherent as an A-module. By Lemma 20229 we see that M is perfect, hence M has a finite free resolution $F_\bullet$ over A. It follows that L is quasi-isomorphic to a finite complex of *free* $A \otimes_R K$-modules. Hence by Lemma 36491 we see that $[L] = n[A \otimes_R K]$ in $K_0(A \otimes_R K)$ for some $n \in \mathbf{Z}$. Applying the map of Lemma 36464 we see that L is trivial. □